

Topological Memory.

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Introduction

Today:

- ▶ Proof.
 1. Walkthrough the math
 2. A
 3. B
- ▶ New experimental results.
- ▶ Answering the code teleportation.

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1 - p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1 - p)) < p$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift rule-and the flip upon majority are local decoders.

Storing Logical State

Logical State

For a random variable X we denote by:

1. $\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p -depolarizing channel.
2. $\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p -depolarized channel followed by an application of local decoder D .

The Theorem

Theorem

Let f be any function with bounded value, denote by $|f|$ it's supremum, Then:

$$\mathbf{E}_{\sim q(t)} [f] \leq \mathbf{E}_{\sim p} [f] + t|f|e^{-\Theta(\sqrt{n})}$$

Lemma

There exists a constant $c > 0$, such:

$$\mathbf{E}_{\sim 2p(1-p)} [f \circ D] \leq \mathbf{E}_{\sim p} [f \circ D] + |f|e^{-c\sqrt{n}}$$

Theorem Proof

By induction on t .

$$\begin{aligned}\mathbf{E}_{\sim q^{(t)}}[f] &= \mathbf{E}_{\sim q^{(t-1)}}[\mathbf{E}_{\sim p}[f \circ D]] \\ &\leq \mathbf{E}_{\sim p}[\mathbf{E}_{\sim p}[f \circ D]] + (t-1)|\mathbf{E}_p[f \circ D]|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim 2p(1-p)}[f \circ D] + (t-1)|f|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-c\sqrt{n}}\end{aligned}$$

When in the last transition we used the lemma.

